ANNEX

DRAFT GUIDELINES FOR THE PREPARATION OF THE REPORTS FROM MEMBER STATES ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION ON OPEN SCIENCE

I. INTRODUCTION

These Guidelines are intended to assist Member States in the preparation of the reports on the implementation of the 2021 Recommendation on Open Science (hereinafter referred to as "the 2021 Recommendation") that was adopted by the 41st session of the General Conference of UNESCO on 23 November 2021.

The 2021 Recommendation proposes seven areas of action: (i) promoting a common understanding of open science, associated benefits and challenges, as well as diverse paths to open science; (ii) developing an enabling policy environment for open science; (iii) investing in open science infrastructures and services; (iv) investing in human resources, training, education, digital literacy and capacity building for open science; (v) fostering a culture of open science and aligning incentives for open science; (vi) promoting innovative approaches for open science at different stages of the scientific process; (vii) promoting international and multi-stakeholder cooperation in the context of open science and with view to reducing digital, technological and knowledge gaps.

For the purpose of the 2021 Recommendation, open science is defined as an inclusive construct that combines various movements and practices aiming to make multilingual scientific knowledge openly available, accessible and reusable for everyone, to increase scientific collaborations and sharing of information for the benefits of science and society, and to open the processes of scientific knowledge creation, evaluation and communication to societal actors beyond the traditional scientific community. It comprises all scientific disciplines and aspects of scholarly practices, including basic and applied sciences, natural and social sciences and the humanities, and it builds on the following key pillars: open scientific knowledge, open science infrastructures, science communication, open engagement of societal actors and open dialogue with other knowledge systems.

Pursuant to Articles 15 and 16.1 of the Rules of Procedure concerning recommendations to Member States and international conventions covered by the terms of Article IV, paragraph 4, of the UNESCO Constitution, the Director-General of UNESCO has invited Member States by the Circular Letter (CL/4381) to submit the 2021 Recommendation to their competent authorities within a period of one year from the close of the session of the General Conference at which it was adopted, i. e. before 24 November 2022.

Furthermore, under Article VIII of UNESCO's Constitution, Member States are required to submit a report on the action taken upon the conventions and recommendations adopted by the General Conference.

II. WHAT ARE THE AIMS OF THIS CONSULTATION?

This global consultation aims to assist Member States in: (1) mapping policies, mechanisms and actions related to open science and all its key pillars, against the objectives of this Recommendation; (2) collecting and disseminating progress and good practices on open science; (3) identifying challenges and opportunities faced by Member States in the implementation of the Recommendation, so as to identify specific capacity-building needs.

III. HOW TO FILL IN THE QUESTIONNAIRE?

The following questionnaire, which was developed with inputs from UNESCO working group on open science monitoring framework, aims to guide and assist Member States with their reporting on the progress made in the implementation of the 2021 Recommendation. It aims to collect information on the extent to which Member States have integrated the core values and guiding principles of open science that are defined in the 2021 Recommendation, in their national science, technology and innovation systems. Responses to this questionnaire will be considered as the official national report of each Member State.

Prior to completing the questionnaire, Member States are encouraged to organize the necessary consultations within and outside the concerned ministries and institutions, including with authorities and bodies responsible for science, technology and innovation, and consult relevant actors concerned with open science, including the scientific community, professional associations, civil society, indigenous and traditional knowledge holders, private sector partners and National Commissions for UNESCO.

In the preparation of their reports, Member States are also encouraged to consult the UNESCO Open Science Toolkit and the first edition of the *UNESCO Open Science Outlook* (published December 2023) containing a collection of information and communication materials relevant to the implementation of the 2021 Recommendation on Open Science. These resources are available online at https://www.unesco.org/en/open-science/toolkit and https://unesdoc.unesco.org/ark:/48223/pf0000387324.

Member States are requested to designate a contact person responsible for information sharing and cooperation with UNESCO in relation to reporting on the 2021 Recommendation.

Member States are encouraged to submit the questionnaire (in English or French) in one of the following ways:

(i) Online (preferred): the consolidated questionnaire (one per Member State) can be completed and submitted online.

(ii) Email: the questionnaire can be completed electronically and sent to: openscience@unesco.org

The responses to the questionnaire, if sent by email, should not exceed 15 pages, excluding annexes and is to be submitted to UNESCO in electronic form only (standard .doc, .pdf or .rtf format).

The reports will be made available on UNESCO's website in order to facilitate the exchange of information relating to the promotion and implementation of this Recommendation.

IV. DRAFT QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: Denmark
- Organization(s) or entity(s) responsible for the preparation of the report*:
  Name: Danish Agency for Higher Education and Science
  Website: www.ufm.dk
  Email address*:
  Phone number*:
  Please describe the role/mandate of your organization: The Danish Agency for Higher Education and Science is an agency in the Ministry of Higher Education and Science and handles tasks within preparation and administration of grants for research, higher education, and research-based innovation, as well as the State Educational Grant and Loan Scheme.
- Officially designated contact person(s) who completed this survey:

Full name*: Hanne-Louise Kirkegaard

Position*: Senior Advisor

Email address*:

- Full Name(s) of designated official(s) certifying the report: Deputy Director General Annemarie Falktoft, Danish Agency for Higher Education and Science.
- Brief description of the consultation process established for the preparation of the report: The Danish universities have been consulted via their organisation: Universities Denmark (the organisation of the eight Danish Universties). Additionally, the following public research funds have been consulted: Independent Research Fund Denmark, Innovation Fund Denmark and Danish National Research Foundation. Finally, DeIC (Danish e-Infrastructure Cooperation) has been consulted.
- Other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey: The Danish National Commission for UNESCO.

  Name of the organization*: The Danish National Commission for UNESCO

  Website*: https://www.unesco.dk/english/the-danish-national-commission-for-unesco-and-the-patron-of-unesco-denmark

  Email address*

  Sector: ◯ Public ◯ Private ◯ Civil Society ◯ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

1.1 Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES / NO

If yes, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so. There are no obstacles to promote the recommendation. The Agency for Higher Education plans to promote the recommendation via the Agency´s website on Open Access/FAIR data in 2025.

1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES / NO

If yes, which language(s)?

1.3 Have there been awareness raising activities on open science, including all its key elements[i], associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? *(ref.: (i) 16.f, 16.g, 16.h)*

YES / NO

If yes, please provide details on the activities organized or foreseen and links as relevant.

In connection with the Ministry of Higher Education and Science's evaluation of Denmark's National Strategy for Open Access, a conference was held on November 1, 2023, with the participation of senior management from Danish research institutions, including universities, university colleges, higher education institutions within the fine arts, other research organizations, as well as public and private foundations. In addition to discussing the implementation of the strategy's goals and recommendations for its revision, the EU Commission gave a presentation on Open Science, including, among other topics, the value of FAIR data.

Furthermore, Denmark, DeiC (Danish e-infrastructure Consortium), is member of the European partnership Knowledge Exchange. Established in 2005, it is a collaboration between six national key organisations in Europe. The purpose is to develop infrastructures and services to enable Open Science for the benefit of research and higher education. Knowledge Exchange outputs have been disseminated to the Danish universities and other relevant stakeholders.

1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open science[ii], in publicly funded research? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

YES / NO

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

The Ministry of Higher Education and Science has established a committee tasked with updating the Danish Code of Conduct for Research Integrity (2014). The committee includes representatives from all Danish higher education institutions, including all universities, public, and private research foundations, as well as representatives of both experienced and early-career researchers. The committee's mandate is to update the code of counduct to reflect developments in Open Science among other things, including guidelines for data management based on FAIR principles, Open Access, and Citizen Science. The code outlines guidelines for good research practices and specifies how responsible research should be conducted in Denmark, including clarifying the division of responsibilities between institutions and researchers.

The update of the code is expected to be completed during the spring of 2025.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy[iii], strategy or plan of action on science, technology and innovation (STI)?

YES / NO

If yes

- please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption,* *year of last update, key areas/sections of the policy, and links to the webpage if available.*

- does it include any of the following open science elements:
  - open access to scientific knowledge
  - open science infrastructure
  - open engagement of societal actors
  - open dialogue with other knowledge systems

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

In your country, is there a policy, or a set of policies and/or legal framework(s)[iv] at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? *(ref.: (i) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h)*

YES / PARTLY /NO

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.*

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.

It is not due to barriers or obstacles as such. Rather, the reason is that the Ministry of Higher Education and Science has chosen to prioritise specific elements of Open Science, including Open Access, FAIR research data, and research integrity, as outlined in Section 2.2.

2.2 In your country, are there specific policy instruments[v] that aim to promote open science?

YES/ UNDER DEVELOPMENT/NO

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

The Danish Ministry of Higher Education focuses on three key elements of Open Science but does not have a national Open Science strategy. The key elements are:

- Open Access to scientific peer-reviewed publications – Denmark's National Strategy for Open Access (2014)
- FAIR research data – National strategy for data management based on the FAIR principles (2021)
- Research Integrity: Danish Code of Conduct for Research Integrity (2014)

Denmark's National Strategy for Open Access aims to provide Open Access to all scientific publications from Danish universities in 2025. Every year, the Danish

Open Access Indicator monitors the progress of Open Access at Danish universities. 2024 monitoring shows that 75 per cent of Danish scientific articles from Danish universities were Open Access. The National Strategy for Open Access was evaluated in 2023-2024. The evaluation shows that the number of scientific articles from Danish universities made freely available online has inceased. However, the strategy's aim of 100% by 2025 is unlikely to be reached due to barriers such as embargo periods and journal restrictions.

National stakeholders such as universities and other research institutions have started implementing the National Strategy for Data Management based on FAIR principles in 2022. Furthermore, a working group of national stakeholders, recently set up by the Danish Ministry of Higher Education and Science, will coordinate measures that require national coordination and monitor the progress of the implementation.

Finally, the Ministry of Higher Education and Science has established a committee tasked with updating the Danish Code of Conduct for Research Integrity (2014) to reflect developments in Open Science among other things, including guidelines for data management based on FAIR principles, Open Access, and Citizen Science, cf. paragraph 1.4.

Links:

- Denmark's National Strategy for Open Access (2014/2018): https://ufm.dk/en/research-and-innovation/cooperation-between-research-and-innovation/open-access/Publications/denmarks-national-strategy-for-open-access/national-strategy-for-open-access-english.pdf

- Evaluation. Denmark's National Open Access Strategy (2024) (Executive summary in English on p 11-14):

  https://ufm.dk/publikationer/2024/filer/evaluering-af-danmarks-nationale-open-access-strategi.pdf

- National strategy for data management based on the FAIR principles (2021): https://www.deic.dk/sites/default/files/documents/PDF/EN_National%20strategi%20for%20data%20management%20baseret%20p%C3%A5%20FAIR-principper.pdf

- Danish Code of Conduct for Research Integrity (2014): https://ufm.dk/en/publications/2014/files-2014-1/the-danish-code-of-conduct-for-research-integrity.pdf

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

2.3 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? *(ref.: (ii) 17.a, 17.b, 17.c, 17.f, 17.h)*

YES / UNDER DEVELOPMENT /NO

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.

Beyond the strategies mentioned in section 2.2., the Ministry of Higher Education and Science has no plans or strategies as mentioned in section 2.3.

2.4 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? *(ref.: (ii) 17.d, 17.e, 17.g)*

YES /UNDER DEVELOPMENT/ NO

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and elements of open science**Fejl! Bogmærke er ikke defineret.** included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

At the institutional level, the eight Danish universities have institutional Open Access and Data Management/FAIR Data policies. Some universities have guidelines regarding Citizen Science. Examples are listed below:

Aarhus University (AU)
At Aarhus University Open Science is addressed in several ways. At the policy level there is an Open Science Committee consisting of a dean (as the chair person), all the vice deans of research, and the vice director of AU Research support. This committee refers directly to the university management and is responsible for the development of all Open Science policies of the university: (https://medarbejdere.au.dk/en/strategy/the-open-science-comittee). Below this Committee, Aarhus University has a Research Data Management policy which addresses a number of Open Science issues: https://medarbejdere.au.dk/en/research-data-management , as well as an Open Access Policy: (https://medarbejdere.au.dk/en/open-access). Moreover, Aarhus University also has a Citizen Science hub to enable knowledge exchange, networking, and collaborations in relation to citizen science: (https://projects.au.dk/citsci)

University of Southern Denmark (SDU)
The SDU Open Science Policy was revised in 2024. The policy is concentrated around FAIR Data and Data Management, Open Access to publications, CoARA, and Citizen Science.

Link to policy: https://sdunet.dk/en/research/research-data-management-support/the-sdu-open-science-policy

Aalborg University (AAU)
AAU has an updated Open Access policy approved in 2024 and in effect from January 2025: (https://www.staff.aau.dk/rules/research/open-access-policy-of-aalborg-university).

AAU also has a datamanagement policy:
(https://www.staff.aau.dk/rules/research/policy-for-research-data-management)

The updated Open Access policy supports AAU's new Research and Innovation indicator, which is CoARA compliant and operates actively with Open Science through a focus on Collaboration, Visibility, and Openness. See:
https://www.researcher.aau.dk/guides/aau-research-indicator and
https://vbn.aau.dk/en/projects/konstruktion-af-forskningsindikator-til-aau

AAU has a new Competence unit and Diamond Publishing platform (AAU OPEN): (https://www.aau.dk/forskning/open-publishing?marketing=open.aau.dk)

Technical University of Denmark (DTU)
DTU has developed a Policy for research data management (https://www.bibliotek.dtu.dk/-/media/andre_universitetsenheder/bibliotek/bibliotek_newdesign/politikker/dtu-research-data-management-policy.pdf) aligned with the National Strategy for datamanagement based on the FAIR principles (https://www.deic.dk/sites/default/files/documents/PDF/EN_National%20strategi%20for%20data%20management%20baseret%20p%C3%A5%20FAIR-principper.pdf).
Perspectives on Opens Access to research outputs are outlined in the DTU Research Publication Policy: (https://www.dtu.dk/-/media/andre_universitetsenheder/bibliotek/Bibliotek_newdesign/politikker/dtu_publication_policy.pdf).

Copenhagen Business School (CBS)
CBS has an Open Access Policy (2019) and Open Access guidelines:
(https://www.cbs.dk/files/cbs.dk/call_to_action/cbs_open_access_policy_2019.pdf).

Furthermore, CBS is currently revising its Research Data Management policy from 2017 (https://www.cbs.dk/files/cbs.dk/call_to_action/cbs_rdm_policy.pdf), which states the following: _"Copenhagen Business School (CBS) envisions itself as a "world-leading business university" that contributes to society with academically excellent research and research-based education. Academic excellence presupposes trustworthiness and high integrity, i.e. honesty, transparency, and accountability of research, as stated in the Danish Code of Conduct for Research Integrity. Research data management ensures honest, transparent, and accountable research practices. Therefore, CBS encourages researchers to manage their data according to the FAIR-principles, thus making them findable, accessible, interoperable, and reusable. CBS and the individual researchers share responsibilities to comply with the principles of responsible and FAIR research data management outlined in this policy. The policy specifies in more detail the obligations of both the institution and the researchers. (§1.1 Purpose)"

Public Foundations
The three public research foundations (Independent Research Fund Denmark, Innovation Fund Denmark and the Danish National Research Foundation) have a joint Open Access policy, which is aligned with the national Open Access policy.: (https://ufm.dk/en/research-and-innovation/cooperation-between-research-and-innovation/open-access/Publications/councils-and-foundations)

Universities
Furtermore, Universities Denmark (the organisation of the eight Danish Universties) has developed a framework for the recognition of knowledge dissemination, which includes guidance on how open science practices can be recognized: (https://dkuni.dk/wp-content/uploads/2024/06/dkuni_advancing_competencies_and_experiences_within_impact.pdf)

2.5 In your country, are there specific funding mechanisms[vi] for open science at the national level? *(ref.: (ii) 17.j, (iii) 18.j, (iv) 19.c)*

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

The Ministry of Higher Education and Science provide funds for a national Open Access Indicator, which monitors the annual progress in Open Access in Denmark, cf. Denmarks National Strategy on Open Access.

Link: The Danish Open Access Indicator: https://oaindikator.dk/en/

Furthermore, DeIC (Danish e-infrastructure cooperation) invests in European Open Science Cloud (EOSC). The EOSC is recognised by the Council of the European Union among the 20 actions of the policy agenda 2022-2024 of the European Research Area (ERA) with the specific objective to deepen open science practices in Europe. It is also recognised as the "science, research and innovation data space" which will be fully articulated with the other sectoral data spaces defined in the European strategy for data.

Link: https://research-and-innovation.ec.europa.eu/strategy/strategy-research-and-innovation/our-digital-future/open-science/european-open-science-cloud-eosc_en

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.

2.6 In your country, have open science practices[vii] been integrated in existing research funding mechanisms?

YES /NO

If yes, please provide details and links as relevant.

The Danish National Research Foundation has introduced various FAIR-related requirements for grant recipients. The foundation regularly approves dedicated data managers (and actively encourages this). Furthermore, Innovation Fund Denmark requires a DMP (Data Mangement Plan) for some of its instruments and funds grant recipients' expenses for Open Access publishing (within reasonable limits).

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? *(ref.: (ii) 17.i)*

YES / NO

If yes, please provide details and links as relevant.

ODIN – an Open Innovation in Science (OIS) platform
For half a decade, Aarhus University has experimented with use-inspired, patent-free, and precompetitive industry collaborations in order to accelerate discoveries and make them easier for industry to translate into innovation. ODIN is an Open Innovation in Science (OIS) platform where industry and academia co-create open research projects relevant for drug discovery and diagnostics. The platform was established by five different universities together with a range of Danish and international companies within pharma and biotech.

OIS platforms foster open and patent-free public-private R&D partnerships in the form of co-created research projects between university researchers and companies. No one can take out patents or other forms of intellectual property rights on the open outputs of OIS platforms. However, anyone can use them for commercial purposes downstream. All research at the platforms must be precompetitive (TRL 1-3) – but also highly use-inspired.

Link: Open ODIN: https://projects.au.dk/open-odin

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

2.7 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy *(ref.: (ii) 17.j, 23.c)*

YES / UNDER DEVELOPMENT / NO

If yes or under development, please provide details and links as relevant.

The Ministry of Higher Education and Science operates a national Open Access Indicator, which monitors the annual progress in Open Access in Denmark, cf. Denmarks National Strategy on Open Access.

Link: The Danish Open Access Indicator: https://oaindikator.dk/en/

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. *(ref.: (iii) 18.a)*

In 2022, Denmark's total R&D expenditure was estimated to 2,89 pct. of GDP, of which 1,11 pct. of GDP was performed in the public sector and 1,78 pct. of GDP was performed in the private sector. Compiled by Statistics Denmark as of the beginning of 2024.

3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. *(ref.: (iii) 18.b)*

In 2024, 97 pct. of the Danish households had access to the internet. Compiled by Statistics Denmark 2024.

3.3 Are there national research and education networks (NRENs)[viii] active in the country? *(ref.: (iii) 18.c)*

YES / NO

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

The responsibility as a Danish NREN is lifted in collaboration between DeiC Forskningsnettet (The Danish Research Network) https://www.deic.dk/en/research-network) and DeiC (Danish e-Infrastructure Consortium).

All open science actors have access to the national research network.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.

3.4 Are there national or institutional open science infrastructures[ix] in your country? *(ref.: (iii) 18.d, 18.e, 18.k, 18.l, 18.m)*

YES / UNDER DEVELOPMENT / NO

If yes or under development, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.

  DeiC Dataverse is a trusted digital repository for metadata and data, i.e. a data management supporting infrastructure, where researchers from Danish research institutions can publish research data.

  Services for data storage and sensitive data storage are under development. The storage service will be linked to DeiC Dataverse.

- community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society

  Denmark is member of several European and international community managed infrastructures. These can be very formal such as European Research Infrastructure Consortiums (ERIC's); The European Strategy Forum on Research Infrastructures (ESFRI) as well as more informal collaborations, e.g. The Global Dataverse Community Consortium (GDCC).

  DeiC is host of a national DataCite consortium that via DataCite international provides Persistent Identifier infrastructure that makes research more effective by connecting research outputs and resources – from data and preprints to images and samples.

- platforms for exchanges and co-creation of knowledge between scientists and society
- community-based monitoring and information systems to complement national, regional and global data and information systems.

  Denmark has via the EOSC EU Node (and coming nodes) access to such information systems, as we are also developing our own national systems. DeiC is mandated member of EOSC Association.

  For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

  - The European Strategy Forum on Research Infrastructures (ESFRI): https://research-and-innovation.ec.europa.eu/strategy/strategy-research-and-innovation/our-digital-future/european-research-infrastructures/eric_en
  - European Research Infrastructure Consortium (ERIC): https://research-and-innovation.ec.europa.eu/strategy/strategy-research-and-innovation/our-digital-future/european-research-infrastructures/eric_en
  - EOSC EU Node: https://open-science-cloud.ec.europa.eu/
  - DataCite: https://datacite.org/
  - The Global Dataverse Community Consortium (GDCC): https://www.gdcc.io/about.html

Furthermore, several OJS publishing platforms ensure Open Access to reserach and allow researchers to publish their work without APCs. See the list below:

- Tidsskrift.dk: https://tidsskrift.dk/
- Rauli.cbs.dk: https://rauli.cbs.dk/
- geus.journals.org: https://www.geusjournals.org/
- journals.aau.dk: https://journals.aau.dk/
- Aalborg University Open Publishing: https://www.en.aau.dk/research/open-publishing/aau-open-publishing
- Openbooks.dk (coming 2025)

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? *(ref.: (iii) 18.e, 18.f)*

YES / NO

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science[i], stage of research, accessibility to all the relevant users in the region or internationally).

Denmark contributes to the development of the European Open Science Cloud (EOSC) via the implementation of the National strategy for data management based on the FAIR principles (2021), cf. paragraph 2.2.

Furthermore, DeIC (Danish e-Infrastructure Cooperation) is currently exploring the possibilities of establishing a Danish or Nordic EOSC node that can be federated with the EOSC EU node as well as other national and thematic nodes in other EU member states.

If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?

YES / NO

3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? *(ref.: (iii) 18.g)*

YES / NO

Denmark is a member of the EU and a number of international research infrastructures (e.g. CERN, ESO, EMBL, ESRF, European XFEL, and ESS). Via those memberships, there are world-leading collaborations on research infrastructures and research with stakeholders in countries from Latin America, Africa, and Asia. The EU as well as all facilites work to promote open science and open access to FAIR data.

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science[i], stage of research, accessibility to all the relevant users in the region or internationally).

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? *(ref.: (iv) 19.a, 19.c)*

YES / NO

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

Capacity building has been focused on ensuring national infrastructure is in place that makes access possible to research data, primarily DeiC Dataverse (https://dataverse.deic.dk/). Furthermore, efforts have been pushed to increase the uptake of Persistent Identifiers for researchers, research data, instruments and more. The Danish universities supports Open Science capacity building with varying amounts of resources and engagement.

In 2025, AAU is launching its Open Science Community Aalborg, which is part of the Skills4EOSC initiative OSC. In Copenhagen, CBS is leading Open Science Community Copenhagen.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

4.2 Have capacity building programmes for policy makers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? *(ref.: (iv) 19.b)*

YES / NO

If yes, please provide more information and links as relevant.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

A framework of open science competencies at a national level is being discussed nationally and by the DeiC board. Support for Open Science competencies is to a varying degree already incorporated locally at the universities into higher education research skills curricula.

National discussions take their point of departure in the National strategy for data management based on the FAIR principles (2021) and The Strategic Research & Innovation Agenda and its Multi-Annual Roadmap (https://eosc.eu/sria-mar/) and specifically in the EOSC project Skills4EOSC (https://www.skills4eosc.eu/).

4.4 Are open educational resources[x] as defined in the 2019 Recommendation on Open Educational Resources, used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? *(ref.: (iv) 19.d)*

YES/ PARTLY/ NO

4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? *(ref.: (iv) 19.e)*

YES / NO

If yes, please provide more information and links as relevant.

The Danish Science Festival is an annual week-long festival that creates meeting points between researchers and citizens. It takes place in April. The festival is comprised of more then 700 lectures and events around the country. The festival attract around 70,000 visitors every year.

Link: https://forsk.dk/kontakt-english/the-danish-science-festival

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? *(ref.: (v) 20.b, 20c, 20i)*

YES / NO

If yes, please provide more information and links as relevant.

Universities Denmark (the organisation of the eight Danish Universties) has developed a framework for the recognition of knowledge dissemination, which includes guidance on how open science practices can be recognized, cf. paragraph 2.4.

Link:https://dkuni.dk/wpcontent/uploads/2024/06/dkuni_advancing_competencies_and_experiences_within_impact.pdf

Furthermore, 1 university college, 7 (out of 8) universities, 3 private research foundations and 2 (out of 3) public research foundations have signed the Agreement on Reforming Research Assessment (ARRA) and thus joined the Coalition on Reforming Research Assessment (CoARA). The Coalition for Advancing Research Assessment (CoARA) is a collective of organisations committed to reforming the methods and processes by which research, researchers, and research organisations are evaluated.

Danish signatories:
https://coara.eu/agreement/signatories/?category%5B0%5D=denmark#signatories
The agreement on Reforming Research Assessment:
https://coara.eu/agreement/the-agreement-full-text/

Aalborg University

In 2022, AAU developed an institutional research and innovation indicator, which is inspired by CoARA principles and is compliant with ARRA. The indicator is used to allocate funding to the departments and is a lever to work actively with Open Science elements within the overall themes of Collaboration, Visibility, and Openness. The indicator promotes diversity in outputs, and research roles as well as applying responsible methods in the analyses of quantitative and qualitative data.
https://www.researcher.aau.dk/guides/aau-research-indicator

Public Research Foundations

One of the public research funds: Independent Research Fund Denmark launched the Fund Strategy 2024-2026: "Independent Research in a World of Change" in the fall of 2024. It includes, among other things, a departure from evaluating researchers solely based on where and how much they have published. This also marks a shift away from researchers' dependency on the powerful scientific publishing houses.

The fund is committed to CoARA, and it works on broader research assessments, for example, by recognizing when researchers contribute valuable datasets and make them accessible to others. The Danish Independent Research Fund has laid out an action plan for 2024-2027, which aims to pave the way for new assessment criteria, where publishing in high-profile journals no longer carries as much weight.

The action plan aims, among other things, to establish a Danish CoARA network, develop a new CV format that allows recognition of diverse roles and career paths, and introduce a new format for publication lists that prioritizes quality and relevance over quantity.

Links: DFF CoARA ACTION PLAN 2024-2027 : https://dff.dk/om-fonden/strategi-og-politikker/dff-coara-action-plan-2024.pdf

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? *(ref.: (v) 20.a, 20.c, 20.d)*

YES / NO

If yes, please provide details and links as relevant.

Please see answers in paragraph **2.4** (Universities Denmark´s framework for the recognition of knowledge dissemination) and **2.6** (ODIN – an Open Innovation in Science (OIS) Platform).

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? *(ref.: (v) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h)*

YES / NO

If yes, for each initiative, please provide more information, including on the stakeholders[xi] that are involved or lead the initiative, and links as relevant.

Please, see answers in paragraph **1.4** (Uddate of the Danish Code of Conduct for research Integrity, paragraph **2.4** (Universities Denmark´s framework for the recognition of knowledge dissemination) and **2.6** (ODIN – an Open Innovation in Science (OIS) Platform) and paragraph **5.1**.(Agreement on Reforming Research Assessment (ARRA) and thus joined the Coalition on Reforming Research Assessment (CoARA)).

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

5.4 Have there been any unintended negative consequences[xii] of open science practices reported in your country? (ref.: (v) 20)

YES / NO

If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.

Denmarks Nationalt Open Access Strategy states : « It is crucial for research and from a socio-economic perspective, that the aggregate public expenditure for scientific publication does not increase significantly as a result of the implementation of Open Access »

Link : Denmark´s National Open Access Strategy page 2: https://ufm.dk/en/research-and-innovation/cooperation-between-research-and-innovation/open-access/Publications/denmarks-national-strategy-for-open-access/national-strategy-for-open-access-english.pdf

According to a recent evaluation of the strategy, the strategy has not succeeded in keeping the estimated publication expenses stable in line with its recommendations. The total estimated publication costs have increased. While expenses for licenses have been kept relatively stable, the costs of Open Access publishing with APC (Article Processing Charges), primarily in gold open-access journals, have risen significantly. This increase cannot be solely attributed to an increase in publication volume. This means that Danish universities are increasingly paying for Open Access publishing, either directly or through their Danish or international co-authors.

Links : Evaluation. Denmark´s National Strategy for Open Access (2024) – Executive summery in English, page 12 : https://ufm.dk/publikationer/2024/filer/evaluering-af-danmarks-nationale-open-access-strategi.pdf

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process? *(ref.: (vi) 21)*

YES, at the national level.

YES, at the institutional level.

YES, at both national and institutional levels.

NO

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.
- promoting open innovation practices.
- Others (with a box to add text)

University of Southern Denmark (SDU)

SDU have established a Citizen Science Knowledge Center, a hub for Open Participatory Practices, that encompass all five faculties as well as the Odense University Hospital. The Center serves as infrastructure at various stages of the research cycle and works with various internal and external partners.

Aalborg University (AAU)

The AAU Research and Innovation indicator awards and merits diversity in research outputs including the use of preprints. Moreover, the indicator awards innovation in research processes as part of the annual strategic performance agreements between the departments and the faculties (see also 5.1.).

https://www.researcher.aau.dk/guides/aau-research-indicator

Technical University of Denmark (DTU)

At DTU they promote and support publication of data by maintaining their own Trusted research data repository – DTU Data It is an essential part of the university RDM policy, that all research projects are documented with a data management plan and that data is published according to the FAIR principles.

Research data management, the FAIR principles and data publication is an integrated part of Responsible Conduct of Research and is mandatory training for all PhD's and new employees.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? *(ref.: (vii) 22.a)*

The Danish Ministry of Higher Education and Education is actively engaged in Open Science initiatives in the EU together with its colleagues from the other EU Member States and Associateted Countries. Denmark has - in line with the other

Member States - endorsed the EU Council Conclusions on Open Science and the European Open Science Cloud (EOSC).

Links:

Council Conclusions on Open Science (2016):

https://data.consilium.europa.eu/doc/document/ST-9526-2016-INIT/en/pdf

Council Conclusions on European Open Science Cloud (2018):

https://data.consilium.europa.eu/doc/document/ST-9029-2018-INIT/en/pdf

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? *(ref.: (vii) 22.b, 22.e, 22.f)*

YES / NO

If yes, please provide links as relevant and more information.

Please see answer in paragraph 7.1.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: *(vii) 22.c*)

YES / NO

If yes, please provide links as relevant and more information.

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES/NO

If yes, please provide more information.

According to the Danish National Commission for UNESCO Open Science aims to make research freely accessible, but its implementation faces challenges. The current publishing system favors established journals with high fees, making Open Access costly for researchers with limited funding. Sharing research data is also complex—quantitative data is easier to share, but qualitative data raises ethical concerns and risks misinterpretation. Additionally, language barriers persist, as English dominates scientific discourse, potentially excluding local communities. To ensure fairness, Open Science should adopt flexible funding models, ethical guidelines for data sharing, and strategies to address linguistic inequalities.

The Ministry of Higher Education and Science agrees that these structural issues are relevant. However, the Ministry would like to draw attention to the fact that Denmark´s National Strategy for Open Access is fundamentally based on a Green Open Access model, where researchers make their articles Open Access by self-archiving them in a digital repository accessible to all. This is free (apart from administrative costs), meaning that researchers do not have to pay a fee to the journal to make their article Open Access. However, the costs of Open Access with APC (Article Processing Charges) payments have still increased significantly in Denmark, cf. section 5.4, despite the strategy's option to achieve Open Access through self-archiving in a repository (green Open Access).

Regarding the sharing of qualitative data, Open Science, including the FAIR principles, does not require all data to be open. If a researcher determines that certain data should remain closed—e.g., because they cannot be anonymized or if data should not be shared for confidentiality or security reasons—it is possible to keep them closed, following the principle: "as open as possible, as closed as necessary."

Finally, concerning the issue of English dominating scientific discourse, the Ministry can point out that it has, in a initial phase (2017-2019), supported Danish scientific journals in their transition to digital formats on the condition that they make their articles Open Access, potentially after an embargo period, cf. Tidskrift.dk (section 3.4).

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i Please refer to the elements of open science defined in the 2021 Recommendation, namely : open scientific knowledge (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), open science infrastructures, open engagement of societal actors and open dialogue with other knowledge systems.

ii The 2021Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the values of open science, namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the guiding principles for open science ,namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.

iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.

iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.

v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).

vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.

vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.

viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.

ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.

x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others (2019 Recommendation on Open Educational Resources)

xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.

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